LCD / LCM SPECIFICATION

WINSTAR Display Co.,Ltd.

華凌光電股份有限公司

SPECIFICATION

MODULE NO.: WG320240BP2

General Specification

Item	Dimension	Unit
Number of dots	320x240	—
Module dimension	160.0 x 109.0 x 13.0 (MAX)	mm
View area	122.0 x 92.0	mm
Active area	115.18 x 86.38	mm
Dot size	0.33 x 0.33	mm
Dot pitch	0.36 x 0.36	mm
LCD type	STN Negative, Blue Transmissive (In LCD production, It will occur slightly color difference. We can only guarantee the same color in the same batch.)	
Duty	1/240	
View direction	6 o'clock	
Backlight Type	LED White	
IC	S1D13700	

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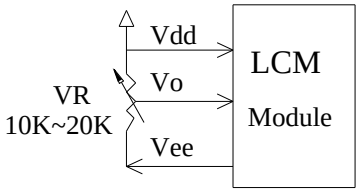
Absolute Maximum Ratings

Item	Symbol	Min	Typ	Max	Unit
Operating Temperature	T_{OP}	-20	–	+70	℃
Storage Temperature	T_{ST}	-30	–	+80	℃
Input Voltage	V_{IN}	-0.3	–	V_{DD}+0.5	V
Output Voltage	V_{OUT}	-0.3	–	V_{DD}+0.5	V
Supply Voltage For Logic	V_{DD}-V_{SS}	0	–	6.5	V
Supply Voltage For LCD	V_{DD}-V_0	0	–	32	V

Electrical Characteristics

Item	Symbol	Condition	Min	Typ	Max	Unit
Supply Voltage For Logic	V_{DD}-V_{SS}	–	4.5	5.0	5.5	V
Supply Voltage For LCD * Note	V_{DD}-V_0	Ta=-20℃	–	–	26.1	V
		Ta=25℃	23.0	23.6	24.2	V
		Ta=70℃	21.8	–	–	V
Input High Volt.	V_{IH}	–	3.5	–	–	V
Input Low Volt.	V_{IL}	–	–	–	1.0	V
Output High Volt.	V_{OH}	–	V_{DD}-0.4	–	–	V
Output Low Volt.	V_{OL}	–	–	–	0.4	V
Supply Current	I_{DD}	V_{DD}=5.0V	90	100	110	mA

* Note: Please design the VOP adjustment circuit on customer's main board



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Interface Pin Function

For 80 family

Pin No.	Symbol	Level	Description
1	V_{SS}	0V	Ground
2	V_{DD}	5.0V	Power supply for Logic
3	V_O	(Variable)	Driving voltage for LCD
4	$\overline{WR}$	H/L	8080 family: Write signal, 6800 family: R/W signal
5	$\overline{RD}$	H/L	8080 family: Read signal, 6800 family: Enable clock
6	CS	H/L	Chip select ,Active L
7	A0	H/L	RD=L WR=H ,A0=L :Data Read AO=H :Status read RD=H WR=L ,A0=L :Data Write AO=H :Command write
8	RES	H/L	Controller reset signal, Active L
9~16	DB0~DB7	H/L	Data bus
17	VEE		Negative voltage output
18	SEL		8088,6800 interface selection (1:68 ,0:80)
19	WAIT		Check busy
20	A		Power supply for B/L +

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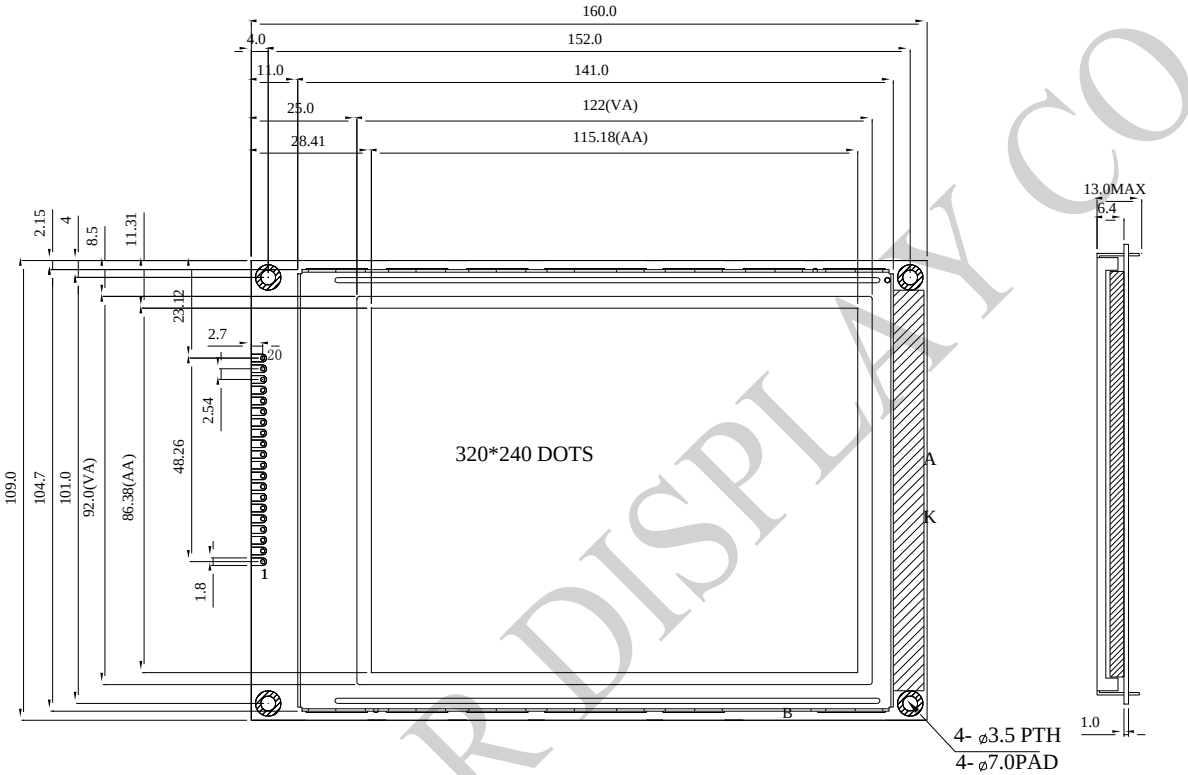
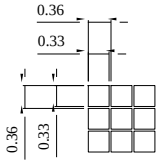
Contour Drawing &Block Diagram





DOT SIZE
SCALE 10/1

PIN NO.	SYMBOL
1	VSS
2	VDD
3	VO
4	WR
5	RD
6	CS
7	AO
8	RES
9	DB0
10	DB1
11	DB2
12	DB3
13	DB4
14	DB5
15	DB6
16	DB7
17	VEE
18	SEL
19	WAIT
20	A

The non-specified tolerance of dimension is ±0.3mm.

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